
Market Concentration and Jurisdictional Dependence in Southeast Asian Internet Infrastructure: Chinese Relative to American Providers, 2024–2026

Jason Hung
Pulse Research Fellow
Primary Contributor

Cecilia Testart
Mentor

Amreesh Phokeer
Advisor

Hanna Kreitem
Advisor

Abstract

China’s Digital Silk Road (DSR) is widely documented, but there remains an absence of empirical measurement of how far Southeast Asian Internet infrastructure depends on Chinese providers. This research study measures the market concentration of content delivery network (CDN) and Internet protocol (IP) infrastructure providers across 10 Southeast Asian countries between 2024 and 2026, drawing on three independently constructed datasets. We computed the Gini coefficient and the Herfindahl-Hirschman Index (HHI) by provider and by jurisdiction, and we estimated three lagged panel regression models on China’s technology investment activities. We found that providers under Chinese jurisdiction hold marginal market shares, never exceeding 3.9 per cent of CDN provision in any country-year, while providers under American jurisdiction hold between 95.9 and 99.6 per cent. Jurisdictional inequality nearly reaches its statistical ceiling in every Southeast Asian market. Compared to CDN provision, IP infrastructure is measurably less market-concentrated, with a non-American and non-Chinese share of between 14.2 and 30.8 per cent. We conclude that Southeast Asian Internet infrastructure dependence is on the United States but not China.

Keywords: *Internet measurement; content delivery networks; market concentration; Internet infrastructure dependence; Digital Silk Road; Southeast Asia*

Introduction

An open, globally connected, secure and trustworthy Internet is a necessary global infrastructure to facilitate international collaboration and exchange, alongside driving social, economic and human development (OECD, 2016). Such an infrastructure is essential to ensure availability, reliability, privacy, data integrity and resilience in digital services (Chaudhary et al., 2022; Laster, 2024). In today's global landscape, the Internet is often dominated by a selected cohort of hyperscalers (meaning large-scale cloud service providers—such as Amazon, Google, and Microsoft—that operate enormous global data centre infrastructures to deliver scalable, high-capacity computing and storage on demand) (Borra, 2024). Given the centralisation of infrastructure, the openness, reliability and resilience of the Internet are subject to multiple challenges. First, the over-reliance on a selected cohort of service providers results in single points of failure (Wilt, 2025). So long as one primary service provider experiences any technical disruption, the global digital economy is at risk of facing a domino effect, demonstrating its systemic fragility (Mantellassi & Paoli, 2024). Second, the service providers have the centralised power to determine how data is optimised, prioritised or filtered before being accessible to users (Hesselman et al., 2020). Such administrative control significantly diminishes user autonomy and, again, challenges the openness and trustworthiness of the Internet (Freuler, 2025).

Since China launched the Digital Silk Road (DSR) initiative in 2015 (Council on Foreign Relations, n.d.), existing literature widely addresses how the country has heavily dominated Internet infrastructure and digital resources at large in the Global South, including Southeast Asian markets (Heeks et al., 2024). Specifically, China has leveraged its DSR initiative to build Internet infrastructure across Africa, Asia, Latin America and beyond (Nouwens, 2021). For example, Chinese service providers, led by Huawei, have been operating globally and investing heavily in telecommunications and 5G technology (Tekir, 2020). Existing literature highlights a few major implications of Chinese service providers' digital and technological dominance. For example, existing arguments raise concerns regarding data privacy, security risks and potential online surveillance by the Chinese Government, as countries worldwide, especially those in the Global South, have increasingly relied on Chinese Internet-technological infrastructure (He & Chen, 2025).

Despite the rich academic, policy and media debates on China's digital dominance across the Global South, especially since the launch of the DSR initiative (e.g., Thussu, 2025), there is a lack of data-driven research specifically addressing the country's dominance in Internet technology across the developing world (Heeks et al., 2024). This project seeks to close that research and data gap, followed by delivering the implications on the openness, reliability, privacy and resilience of the Internet. Among the available Internet measurement metrics, we specifically focus on extracting data from various sources to investigate (the lack of) market concentration of Chinese content delivery network (CDN) and Internet protocol (IP) infrastructure providers across Southeast Asia by the corresponding regional countries' market shares. We further study the market shares of service providers under Chinese jurisdiction, relative to those under the American counterpart. Existing scholarship argues that the United States has continued to lead in Internet imperialism, while China's Internet infrastructure market shares have soared in recent decades and the major Western European counterparts have declined (Winseck, 2017).

Therefore, we comparatively analyse how Southeast Asian markets are concentrated by Chinese

Internet infrastructure providers relative to their American counterparts. We specifically assess Southeast Asian markets because the region is geopolitically important for both the United States and China owing to the maritime trade routes, strategic competition in the Indo-Pacific and economic influence (Vo et al., 2023). It is well-documented that Hong Kong, Taiwan and Singapore have long been maintaining close economic partnerships with mainland China (e.g., Fook, 2018; Bolt, 1996). Comparatively, other Southeast Asian countries' digital economic partnerships with China, especially in the Internet infrastructure domain, are underexplored. Therefore, we study Southeast Asian markets exclusively, with the acknowledgement that the empirical findings and associated geopolitical implications may be applicable to broader regional contexts.

The Internet Society (ISOC) Pulse (n.d.) research team has tracked Internet service concentration since 2021. Their outputs indicate that market concentration among CDN providers, measured by Herfindahl-Hirschman Index (HHI), is by far the most severe among all studied Internet measurement metrics. Additionally, we acknowledge that the measurement of CDN provision alone does not fully address Southeast Asia's Internet infrastructure dependence. When Southeast Asian users browse a website, content may not necessarily be served from a nearby CDN provider. Content can instead be served directly from the original server. In order to comprehensively assess which jurisdiction's Internet infrastructure such traffic depends on, we need to study the IP address and the belonging Autonomous System number (ASN), too. Therefore, this research paper exclusively examines CDN and IP infrastructure providers and the corresponding jurisdictions when evaluating Southeast Asian Internet infrastructure dependency.

Research Aim, Objectives and Questions

The overarching research aim of this study is to quantify and analyse the market concentration of Chinese CDN and IP infrastructure providers across Southeast Asia, especially relative to that of the American counterparts. The first research objective is to map and identify the jurisdictional representation of the corresponding Internet service providers. The second research objective is to quantify and measure the Gini coefficient and HHI to calculate the severity of market concentration (by jurisdiction) in Southeast Asian markets. The third research objective is to study any association between bilateral DSR activity levels and Chinese Internet infrastructure market concentration in each Southeast Asian country. The following are our research questions (RQs):

- 1) **RQ1:** To what extent are Chinese Internet infrastructure providers (i.e., CDN and IP infrastructure) concentrated in Southeast Asian markets as measured by the Gini coefficient?
- 2) **RQ2:** What are the HHI scores for Internet infrastructure market concentration across Southeast Asia?
- 3) **RQ3:** Are stronger DSR ties with China associated with higher degrees of Internet infrastructure dependency across Southeast Asia?

Methods

Data Sources

We employed a mixed-methods approach in this study. We used three panel datasets to measure CDN and IP infrastructure providers serving Southeast Asian visitors. The first dataset was shared

internally by A.P. from the ISOC. Its measurement framework is fully documented in Madeley et al. (2024). In brief, the framework was designed to retrieve the top 1,000 most popular websites per country from the Google Chrome User Experience Report (CrUX), sanitise the list of censored websites using the CitizenLab test lists, and scan the remaining domains from residential proxy vantage points within the country of interest. In this study, we extracted data for Southeast Asian countries only. The ISOC dataset covers June 2024 to January 2026.

We constructed the second dataset in two stages. In the first stage, we queried the CrUX on BigQuery to retrieve the 1,000 most visited website domains in each Southeast Asian country between January 2024 and May 2026. We applied the *SELECT DISTINCT* clause to ensure that no duplicated but separately indexed URL domains appear in our dataset. In the second stage, we identified the CDN provider serving each of these domains. We evaluated several candidate CDN look-up tools and methods, four of which we rejected on methodological grounds. HTTP Archive’s crawl.pages table, despite being filterable to CDN technologies (in category 31) via Wappalyzer detections, crawls exclusively from United States-based Google Cloud regions. United States-based crawl results are not representative of the content delivery experienced by Southeast Asian users. Running the HTTP Archive fork of Wappalyzer locally, in addition, was infeasible. This is because Wappalyzer is closed source as of conducting this research. Without open-access methodologies, we cannot verify that we can reliably use this tool for CDN look-ups. We, furthermore, tried to use CRFT Lookup. However, its fingerprint database is not publicly auditable. Moreover, we excluded CDN Finder because its crawls originate from a Singaporean data centre alone and, beyond Cloudflare, often return unknown CDN look-up results. We eventually decided to use the open-source Nedko CDN Checker (<https://www.nedko.info/cdn-check/>), whose source code is publicly available and auditable and whose detection logic relies on HTTP headers, DNS CNAME records and server response patterns that align with the data extraction strategy adopted by CDN Planet-owned CDN Finder. As this Nedko CDN Checker is a Python Flask application, we cloned and ran it locally from a residential connection in Bangkok and VPN endpoints in all other Southeast Asian capital cities.

The first two datasets were used to measure CDN providers and the associated jurisdictions. We further compiled the third dataset exclusively to identify and measure IP infrastructure. We built the dataset via OpenINTEL forward-DNS measurements of the Google CrUX top lists for all Southeast Asian countries, covering March 2025 to June 2026. OpenINTEL systematically queries and archives a fixed set of DNS records for each fully qualified domain, including A records (mapping a domain name to an IPv4 address), AAAA records (mapping a domain name to an IPv6 address), NS records (authoritative name servers), alongside SOA, MX, TXT and cryptographic DNSSEC records. These records allow each domain to be mapped to an IP address and its corresponding ASN. Compiling this dataset enables us to attribute the IP infrastructure underpinning each popular domain browsed by Southeast Asian visitors to a specific provider. In the next sub-section, we detail how, upon compiling datasets with lists of providers, we identified the corresponding jurisdictions.

Operationalisation and Measurement: Provider Identification and Jurisdiction

To date, there is an absence of universally agreed definition and categorisation of Internet infrastructure providers by jurisdiction (e.g., Yin et al., 2015). We designed and applied two multi-staged procedures, the first to identify the jurisdiction of each CDN provider featured in the ISOC

and Nedko datasets and the second to categorise the jurisdiction of each IP infrastructure provider recorded in the OpenINTEL dataset.

We began the CDN procedure by compiling a provider inventory. In Stage 1, we extracted all unique CDN provider labels from both the ISOC and the Nedko datasets and pooled them into an Excel spreadsheet. On the spreadsheet, there are a total of 27 CDN providers, of which 11 appear in the Nedko dataset alone, 10 in the ISOC dataset alone and six in both. In Stage 2, we identified the legal name of each CDN provider. We manually searched the official company website via Google Search. We then manually reviewed the corporate documentation published on the website, such as the terms of service, the legal notice and the contact page, in order to record the actual legal name of the operating entity under which the service is marketed. In Stage 3, we took parent companies into account. Where the documentation states that the entity is a subsidiary of a named parent, we took and recorded the parent company’s legal name, so that the jurisdiction subsequently identified refers to the controlling entity and not the local operating entity. In Stage 4, we verified each legal name against a public registry. We searched every legal name in OpenCorporates, where the company record reports the corresponding jurisdiction of incorporation and the registered address. In every manual search on OpenCorporates, the jurisdiction of incorporation and the registered address of the company record referred to the same country, which cross-validates the jurisdiction of the controlling entity of any given CDN provider.

We applied a different multi-staged procedure to identify the jurisdiction of a given IP infrastructure provider from the OpenINTEL datasets. In Stage 1, we retrieved the ASN. The OpenINTEL forward-DNS measurements yield the IP address and the ASN serving each domain. The ASN identifies the network operator carrying that traffic. In Stage 2, we mapped ASNs to organisations using the Center for Applied Internet Data Analysis (CAIDA) AS-Organisations dataset, applied longitudinally on a monthly basis (CAIDA, n.d.). We downloaded the CAIDA monthly snapshots covering March 2025 to June 2026 and matched every row of the OpenINTEL dataset to the CAIDA snapshot of the same measurement month. In Stage 3, we recorded the IP infrastructure provider name. Each ASN retrieved from the OpenINTEL datasets was matched to the registered organisation in the CAIDA records, and that organisation name was recorded as the provider serving the domain. Given the volume involved, the organisation names, covering 721 unique IP infrastructure providers in the OpenINTEL datasets, were read programmatically but not manually from the CAIDA records. In Stage 4, we recorded the jurisdiction. We took the country reported for that same organisation in the CAIDA records as its jurisdiction (i.e., the country in which the organisation registered the ASN with its regional Internet registry).

In sum, the CDN procedure relies on the jurisdiction of incorporation of the controlling legal entity, verified against the company record on the OpenCorporates database. The IP infrastructure procedure, alternatively, relies on the country under which an organisation registered its network resources with a regional Internet registry, which is a proxy for jurisdiction.

Statistical Data Analysis

To answer RQ1, we computed the Gini coefficient to measure how unevenly CDN provision and IP infrastructure market shares are distributed across (1) individual providers and (2) providers aggregated by jurisdiction in each Southeast Asian country. For each country and each Internet infrastructure metric, we calculated the market share of every provider within each country-month,

since the three datasets record different measurement units and cannot share a single definition. In the Nedko dataset, a CDN provider’s share is the proportion of the top 1,000 most visited domains it is detected as serving. The ISOC dataset does not list individual websites and instead reports a count of websites served by each CDN in each country on each crawl date, so a provider’s weight is the sum of its locally hosted and externally hosted website counts. Because the raw ISOC file repeats the same count once per observing vantage point, we retained one record per country, month and CDN, and where a month contains more than one crawl date we took the latest date in that month as the monthly reading. OpenINTEL records each website once a day, so over a month the same website reappears about 30 times. Each website is also listed separately for every IP address it resolves to, and one website can resolve to several IP addresses. To keep from counting the same website many times over, we reduced these repeated records so that, within a single country and month, each website is counted once for every serving IP infrastructure provider. A website served by two providers therefore adds one to each of those two providers. Each country-year figure reported in this paper is the mean of the corresponding monthly figures within that year. The Gini coefficient is computed as:

$$1) \ G = (\sum_i \sum_j |s_i - s_j|) / (2n^2\bar{s})$$

where s_i and s_j denote the market shares of providers i and j , n denotes the total number of providers identified in the corresponding country market, and $\bar{s}$ denotes the mean market share. The Gini coefficient ranges from 0 (meaning perfect equality, where all providers hold identical market shares) to 1 (meaning maximal inequality, where a single provider serves all measured domains). It is noteworthy that, with a finite number of units n , the attainable maximum is $(n - 1)/n$ instead of exactly 1. We treated jurisdiction as a three-category variable, representing the market of China, the United States and Other. For three categories the maximum is 0.667 ($= (3 - 1)/3$), so a jurisdictional Gini near 0.667 signals that one jurisdiction dominates the market almost completely.

To answer RQ2, we computed the HHI, the standard measure of market concentration adopted by the ISOC Pulse (n.d.) research team, for both the CDN and IP infrastructure markets in each Southeast Asian country. We calculated the HHI scores across (1) providers and (2) with providers aggregated by jurisdiction in each Southeast Asian country. The HHI is computed as:

$$2) \ HHI = \sum_i s_i^2$$

where s_i denotes the percentage market share of unit i , which is an individual provider or a jurisdiction depending on which of the two measurements is being computed. The HHI ranges from near 0 to 10,000. According to the ISOC Pulse (n.d.), an HHI score below 1,500 is considered low market concentration; between 1,500 and 2,500 refers to moderate market concentration; between 2,500 and 5,000 is deemed high market concentration; and above 5,000 is regarded as extremely high market concentration.

To answer RQ3, we built three linear panel regression models (with lagged explanatory variables). We estimated the CDN outcome twice, once on the Nedko dataset and once on the ISOC dataset, so that the two independently constructed CDN provider measurements allow cross-checking. Data for China’s DSR investment activities were drawn from the American Enterprise Institute’s China Global Investment Tracker (CGIT), filtered to the technology sector, for the period 2023–2025. From the CGIT, we constructed three explanatory variables, namely $CGIT_projects_{i,t-1}$ (referring

to the number of Chinese-invested or Chinese-constructed technology projects recorded in any given Southeast Asian country in any given year), $CGIT_amount_{i,t-1}$ (referring to the total transaction amount (in millions of US\$) of these projects), and $CGIT_type_{i,t-1}$ (treated as a dummy variable coded 1 where the country-year contains at least one construction transaction and 0 otherwise). The dependent variables are $CDN_China_{i,t}$ (referring to the market share of CDN providers under Chinese jurisdiction in country i in year t , available annually for 2024–2026), and $IP_China_{i,t}$ (referring to the corresponding market share of IP infrastructure providers under Chinese jurisdiction).

It is noteworthy that the OpenINTEL forward-DNS measurements of the Google CrUX top lists are only available from March 2025 onwards. Therefore, Models 1a and 1b are estimated on a three-year panel ($T = 3$; years: 2024, 2025, 2026), with the explanatory variables lagged to 2023, 2024 and 2025, respectively. Model 2 is estimated on a two-year panel ($T = 2$; years: 2025, 2026), with the explanatory variables lagged to 2024 and 2025, respectively. To construct each annual observation in a transparent manner, we averaged the monthly measurements available within that calendar year.

All explanatory variables are lagged by one period (year $t - 1$) to mitigate issues of autocorrelation and to avoid running spurious regressions with false significance. We further clustered standard errors at the country level to make our estimates robust to heteroscedasticity and autocorrelation within each country's short time series. We specified two models, each estimated by both fixed-effects and random-effects estimators. Below are the regression equations and assumptions:

Model 1: Chinese CDN provision market concentration

$$3) \quad CDN_China_{i,t} = \beta_0 + \beta_1 CGIT_projects_{i,t-1} + \beta_2 CGIT_amount_{i,t-1} + \beta_3 CGIT_type_{i,t-1} + \alpha_i + \epsilon_{i,t}$$

Model 2: Chinese IP infrastructure market concentration

$$4) \quad IP_China_{i,t} = \beta_0 + \beta_1 CGIT_projects_{i,t-1} + \beta_2 CGIT_amount_{i,t-1} + \beta_3 CGIT_type_{i,t-1} + \alpha_i + \epsilon_{i,t}$$

Under the fixed-effects specification, α_i represents the unobserved time-invariant country-specific effects (group fixed-effects), absorbing all time-invariant, country-specific differences into the intercept, and effectively estimating only the within-country variation. $\epsilon_{i,t}$ is the idiosyncratic error term, assumed to be independent across observations but heteroscedastic (i.e., not identically distributed): $E[\epsilon_{i,t} | X_{i,t}, \alpha_i] = 0$ and $Var(\epsilon_{i,t}) = \sigma_{i,t}^2$.

Under the random-effects specification, α_i is instead modelled as random and assumed to be uncorrelated with the regressors, allowing for estimation using both within- and between-country variation, with the composite error term $u_{i,t} = \alpha_i + \epsilon_{i,t}$. In both model specifications, the use of clustered standard errors at the country level assumes that the errors may be correlated within each country (i.e., autocorrelation) and may have different variances across countries (i.e., heteroscedasticity), but are independent across countries (i.e., clusters).

To choose between the fixed-effects and random-effects estimators for each model, we conducted the Hausman test. The null hypothesis (H_0) of the Hausman test suggests to use the random-effects model as the preferred and most efficient estimator. We conducted the Hausman test on the difference between the fixed-effects and the random-effects coefficient vectors in each model. It is noteworthy that the Hausman statistic is not guaranteed to be non-negative in finite samples. In Model 2, the Hausman statistic returned a negative value, in which case the test is not statistically informative, and we accordingly report both (fixed-effects and random-effects) estimators side by side. We acknowledge that, given the short panels ($T = 3$ for Model 1; $T = 2$ for Model 2), we cannot meaningfully run robust panel unit root tests, as these methods require longer time spans. The two-year panel used to build Model 2 limits the within-country variation available to the fixed-effects estimator. However, any potential spurious regression concerns should be minimised given our methodological design, as our dependent variables are bounded concentration measures and our lagged explanatory variables with clustered standard errors discourage autocorrelation-driven false significance. All data processing and statistical analyses were performed in Python 3, using *pandas* and *numpy* for the panel construction, *linearmodels* for the fixed-effects and random-effects estimation, *scipy* for the Hausman test and *matplotlib* for the figures. The full Python-based pipeline, along with the datasets built and used and the empirical findings visualised, is publicly deposited on a public GitHub repository at <https://github.com/newlivehung123123/pulse> to optimise transparency, reproducibility and replicability. It is noteworthy that, statistically speaking, this research paper reports associations but not causal relationships in answering RQ3.

Results

Concentration by provider (RQ1)

Figure 1 shows the Gini coefficient of provider concentration in the Nedko CDN data by country and month, January 2024 to May 2026. All 10 series (each representing one Southeast Asian country) lie between 0.723 and 0.846. We can see that Laos and Cambodia are the most unequal markets, while Singapore is the least unequal in every month. Also, no Southeast Asian country trends downwards in inequality. Among all Southeast Asian markets, we can see that the market share of Cloudflare in CDN provision grows from between 44.3 and 62.1 per cent in 2024 to between 48.9 and 67.4 per cent in 2026. Alibaba Cloud CDN holds between 0.3 and 1.1 per cent in market share in 2026 and Tencent Cloud CDN never exceeds 0.3 per cent in any country-year (Appendix 1).

Figure 1: Gini of provider concentration, Nedko (CDN), per country over time.

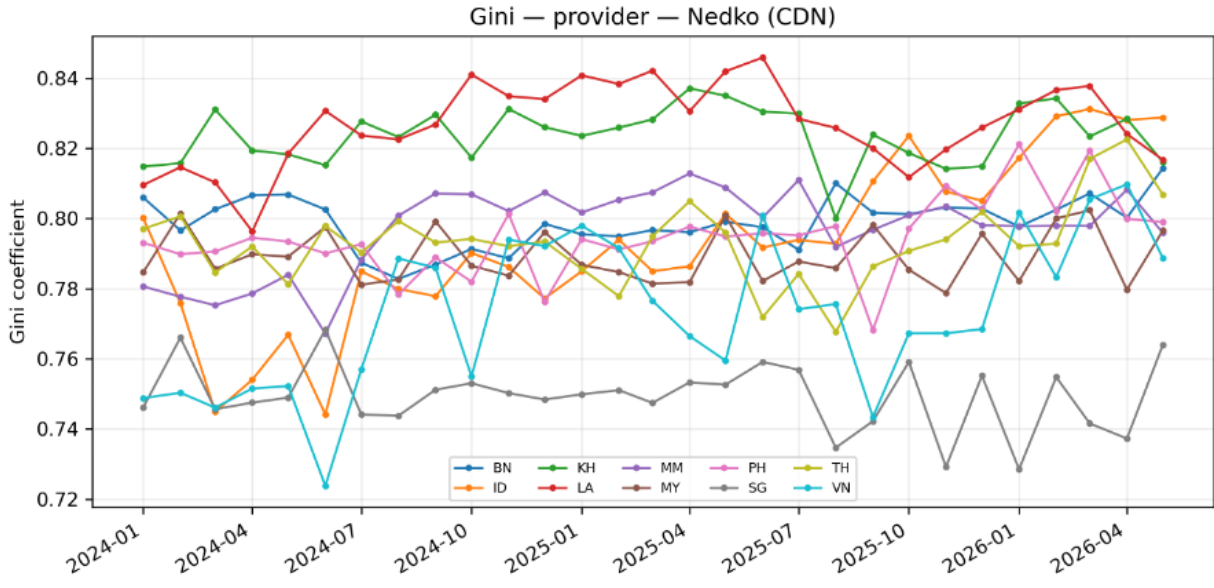


Figure 2 shows the Gini coefficient of provider concentration in the ISOC CDN data by country and month, June 2024 to January 2026. We can see that Thailand and Laos are the most unequal while Singapore, again, is the least. Brunei and Vietnam sitting at approximately 0.28 are clearly a coverage artefact, and their measurements recover in the following month. As in the Nedko CDN data, Cloudflare leads at between 58.6 and 83.0 per cent in market share within Southeast Asian markets in 2026, and Alibaba is the only CDN provider registered in a Chinese jurisdiction (reaching as high as 2.4 per cent in the Cambodian market), according to the ISOC CDN data (Appendix 2).

Figure 2: Gini of provider concentration, ISOC (CDN), per country over time.

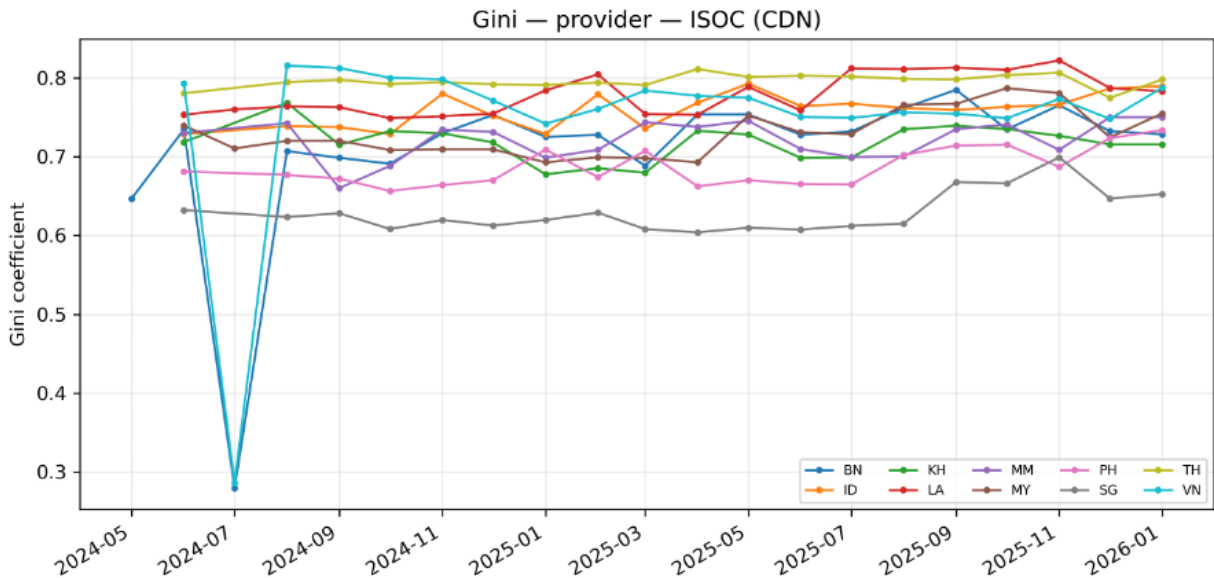
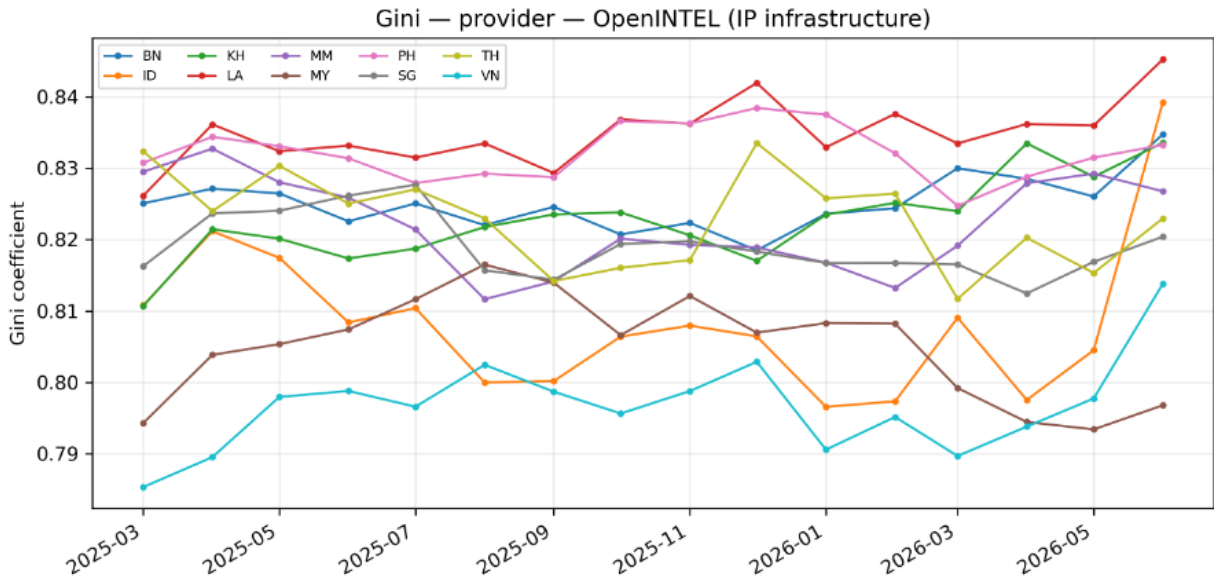


Figure 3 shows the Gini coefficient of provider concentration in the OpenINTEL data by country and month, March 2025 to June 2026. The 10 series share very similar Gini coefficients, between

0.785 and 0.845. We can see that Laos and the Philippines have the highest Gini coefficients and Vietnam has the lowest. Appendix 3 shows the 12 largest IP infrastructure providers (while the rest are counted as Others) in each Southeast Asian market. None of the 12 largest IP infrastructure providers in any given Southeast Asian market belongs to Chinese jurisdiction. Once again, in the OpenINTEL data, Cloudflare holds between 42.9 and 60.7 per cent in market share in 2026 (Appendix 3).

Figure 3: Gini of provider concentration, OpenINTEL (IP infrastructure), per country over time.



Concentration by jurisdiction (RQ1)

Figure 4 shows the Gini coefficient of jurisdictional concentration in the Nedko CDN data. As mentioned, when the market is divided into three fixed categories, the Gini coefficient cannot exceed 0.667. We found that all 10 series lie between 0.634 and 0.661 (which is between 95 and 99 per cent of that ceiling). Vietnam has the highest Gini coefficients and Singapore has the lowest. The American market share drops in all 10 Southeast Asian markets between 2024 and 2026. Also, the Chinese market share never exceeds 1.6 per cent in any country-year (Appendix 4).

Figure 4: Gini of jurisdiction concentration, Nedko (CDN), per country over time.

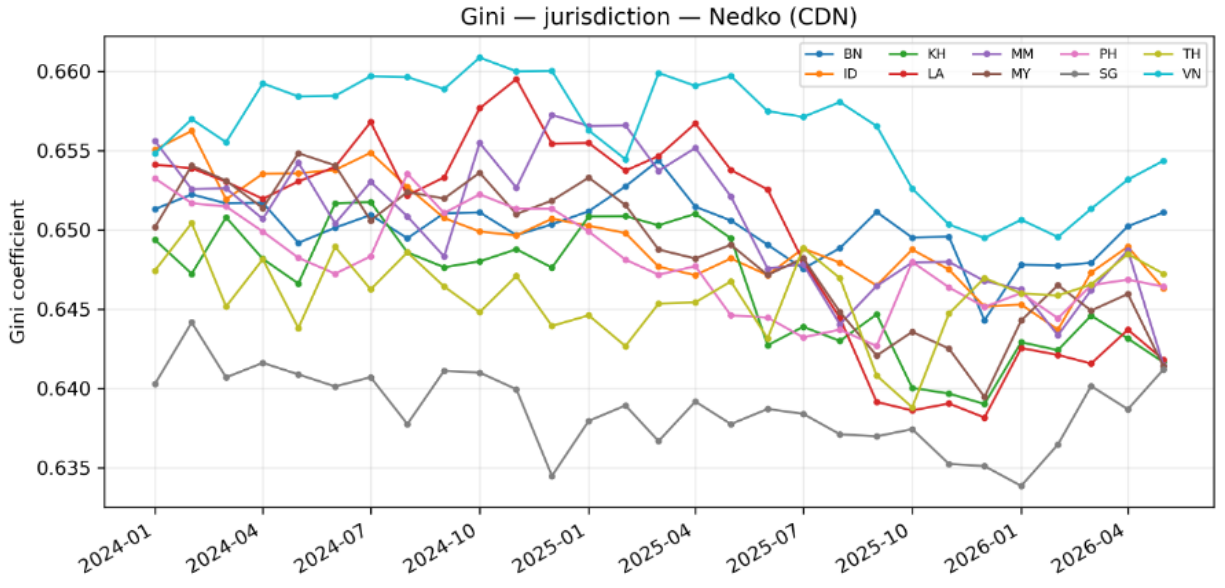


Figure 5 shows the Gini coefficient of jurisdictional concentration in the ISOC CDN data. We can see that all 10 series lie between 0.630 and 0.667. Myanmar is the least concentrated market in almost every month. Myanmar also holds the largest Chinese market share in this study at 3.9 per cent in 2024. The Chinese market share in Myanmar falls to 2.1 per cent by 2026 while its American market share rises from 95.9 to 97.7 per cent between 2024 and 2026 (Appendix 5).

Figure 5: Gini of jurisdiction concentration, ISOC (CDN), per country over time.

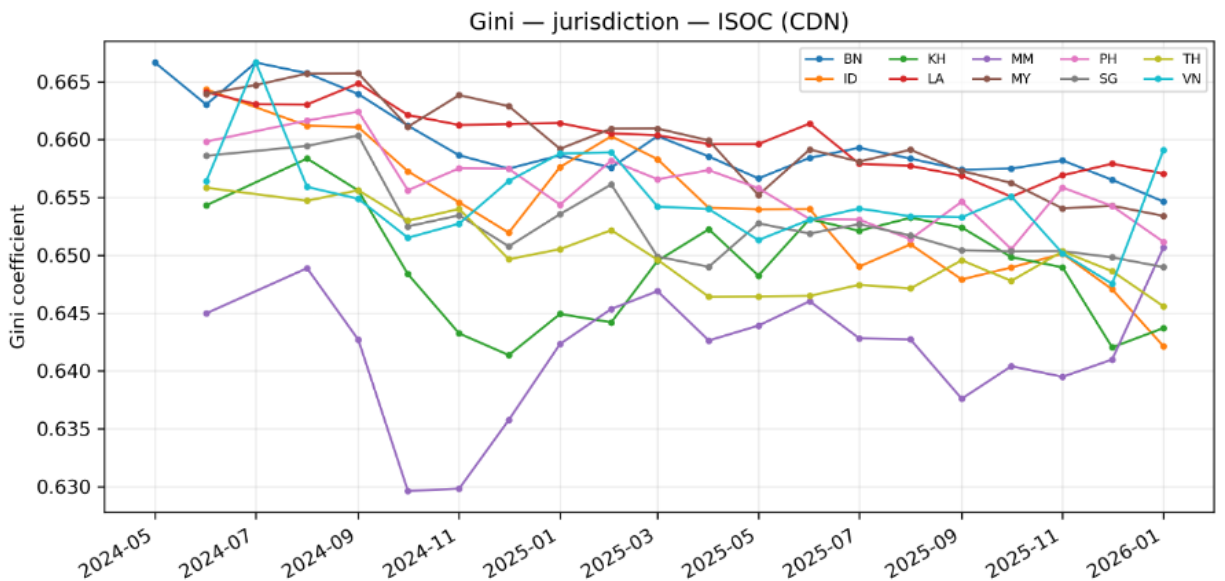
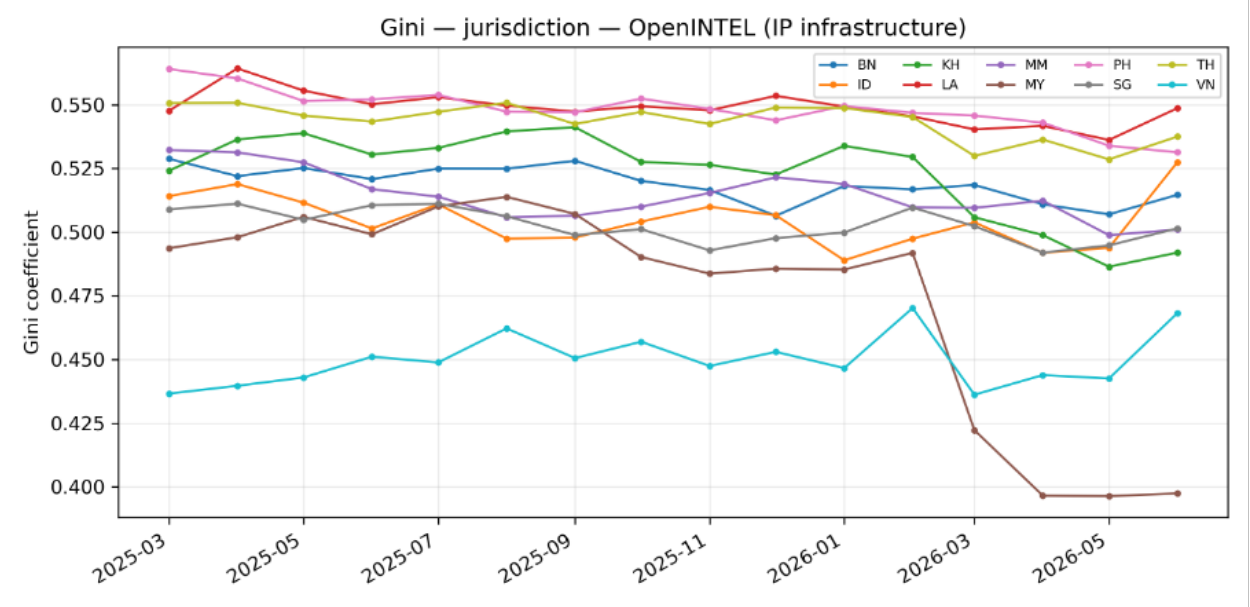


Figure 6 shows the Gini coefficient of jurisdictional concentration in the OpenINTEL IP-infrastructure data, where values sit between 0.397 and 0.565. The Philippines, Laos and Thailand are the most concentrated markets and Vietnam the least. Malaysia falls from approximately 0.492 in February 2026 to 0.397 in April 2026 and ends below Vietnam by June 2026. For the Malaysian

market, the Chinese share rises from 1.3 to 4.8 per cent and the American share falls from 76.1 to 69.5 per cent (Appendix 6).

Figure 6: Gini of jurisdiction concentration, OpenINTEL (IP infrastructure), per country over time.



The intensity of concentration (RQ2)

Figure 7 shows the HHI of provider concentration in the Nedko CDN data. All HHI scores sit between 3,055 and 5,460, so every country-month is at least highly concentrated. Singapore is lowest in terms of its HHI scores throughout the measurement period. Also, Laos exceeds 5,000 at around December 2024 as an extremely concentrated market and peaks at 5,460 in April 2025 (Appendix 1).

Figure 7: HHI of provider concentration, Nedko (CDN), per country over time.

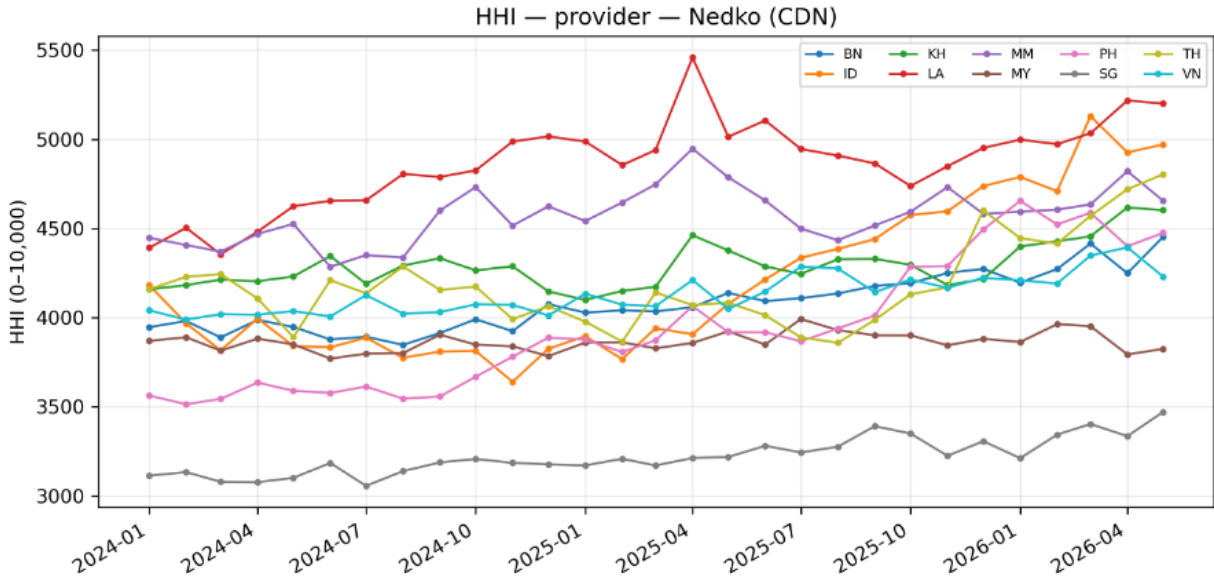


Figure 8 presents the HHI of provider concentration in the ISOC CDN data. All HHI scores sit between 3,280 and 7,280. A total of eight out of 10 Southeast Asian countries sit above 5,000 for most of the measurement period. Singapore stays below 4,200 throughout the period. The Philippines, in addition, crosses 5,000 in around November 2025 (Appendix 2).

Figure 8: HHI of provider concentration, ISOC (CDN), per country over time.

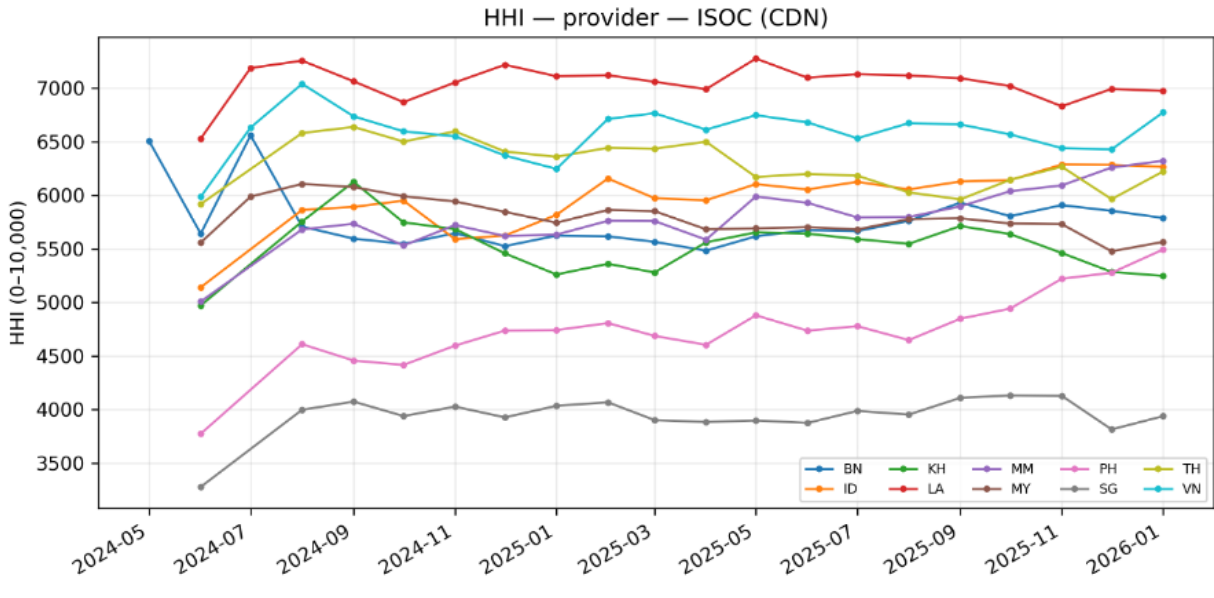


Figure 9 shows the HHI of provider concentration in the OpenINTEL IP infrastructure data, where values range from 1,800 to 4,060. No country reaches 5,000 in any month. Singapore is moderately concentrated throughout the measurement period, and Malaysia falls from approximately 2,500 in February 2026 to 1,800 by April 2026 (Appendix 3). Compared with the HHI scores derived from the CDN data (Figures 7 and 8), Figure 9 shows that provider concentration in IP infrastructure is

substantially lower across Southeast Asian markets. The provider concentration, however, remains high in six of the 10 countries.

Figure 9: HHI of provider concentration, OpenINTEL (IP infrastructure), per country over time.

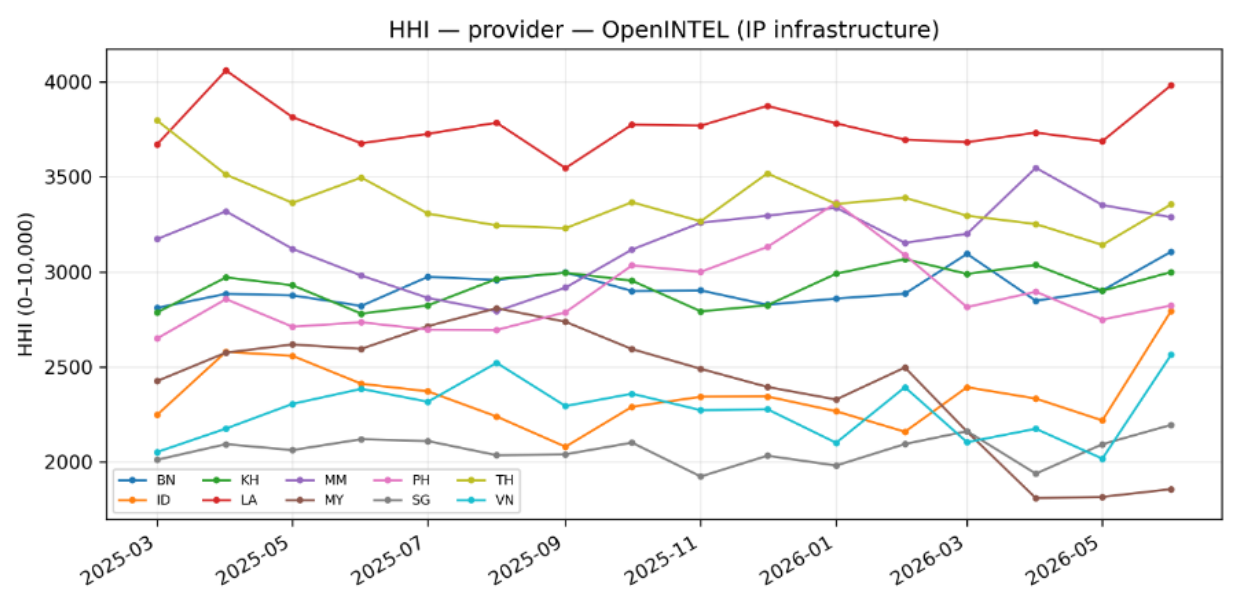


Figure 10 shows the HHI of jurisdictional concentration in the Nedko CDN data. The HHI in every country and month lies between 9,338 and 9,855. We can see a common trend among many Southeast Asian countries in which the HHI trajectories decrease moderately from mid-2025, from roughly 9,700 to around 9,500. Overall, jurisdictional concentration across Southeast Asia remains at an extremely high level (Appendix 4).

Figure 10: HHI of jurisdiction concentration, Nedko (CDN), per country over time.

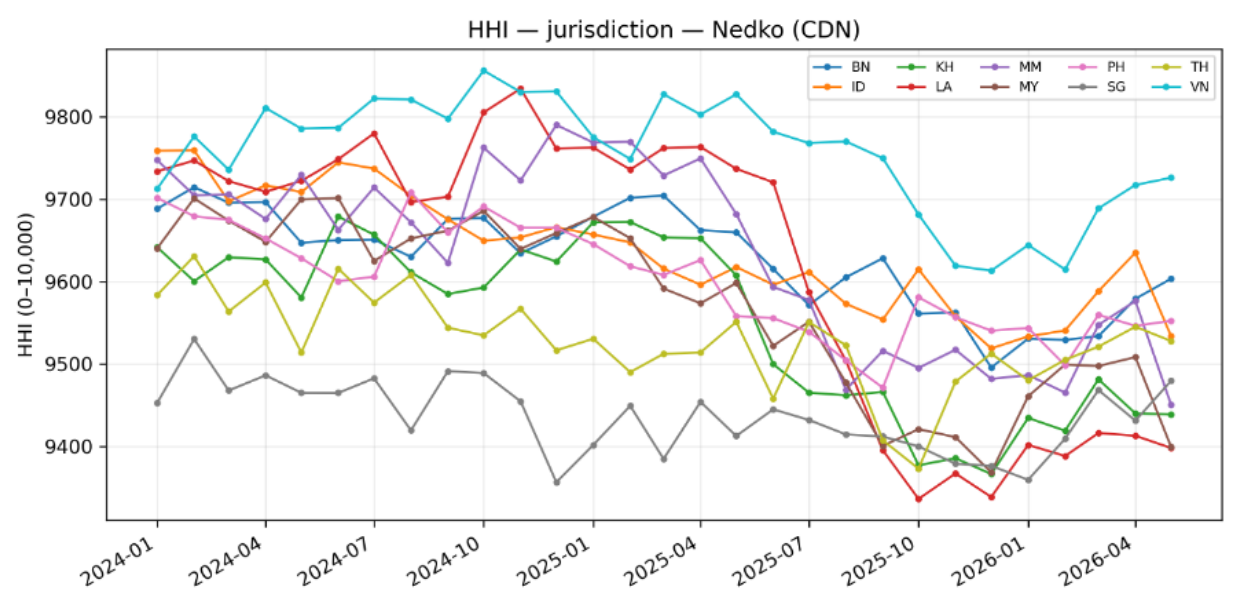


Figure 11 shows the HHI of jurisdictional concentration in the ISOC CDN data, where values range from 8,950 to 10,000. We see that Myanmar is the least concentrated market regionally. Brunei records the maximum of 10,000 in May 2024, and Brunei and Vietnam both record 10,000 in July 2024, indicating that every classified CDN provider serving these countries in those months was registered in the United States (Appendix 5).

Figure 11: HHI of jurisdiction concentration, ISOC (CDN), per country over time.

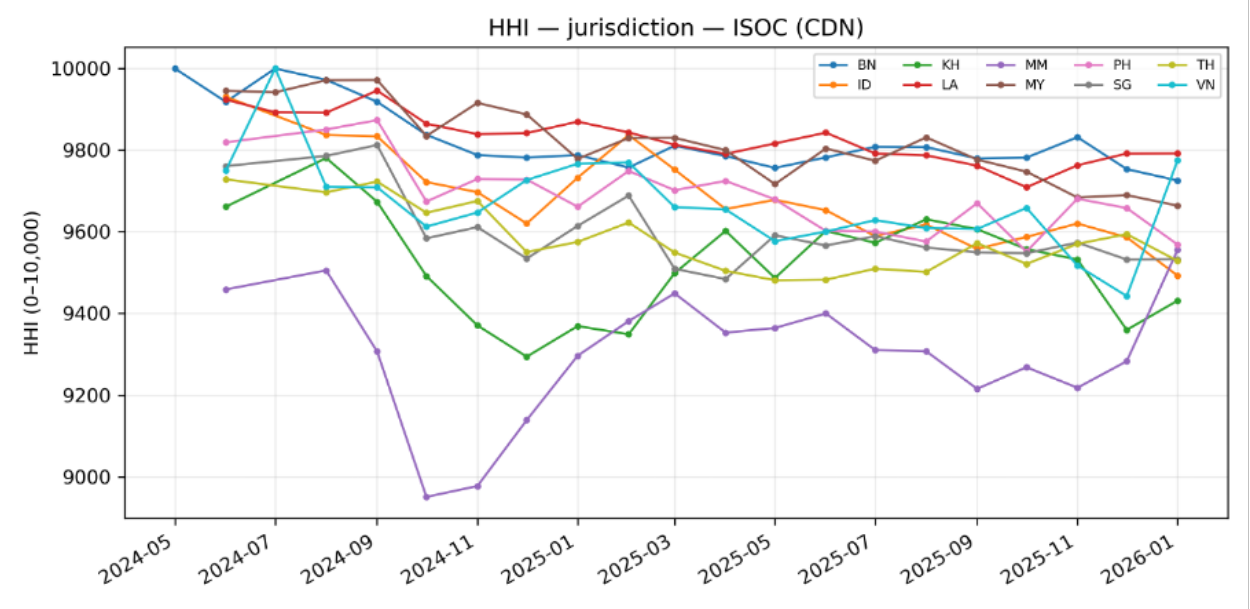
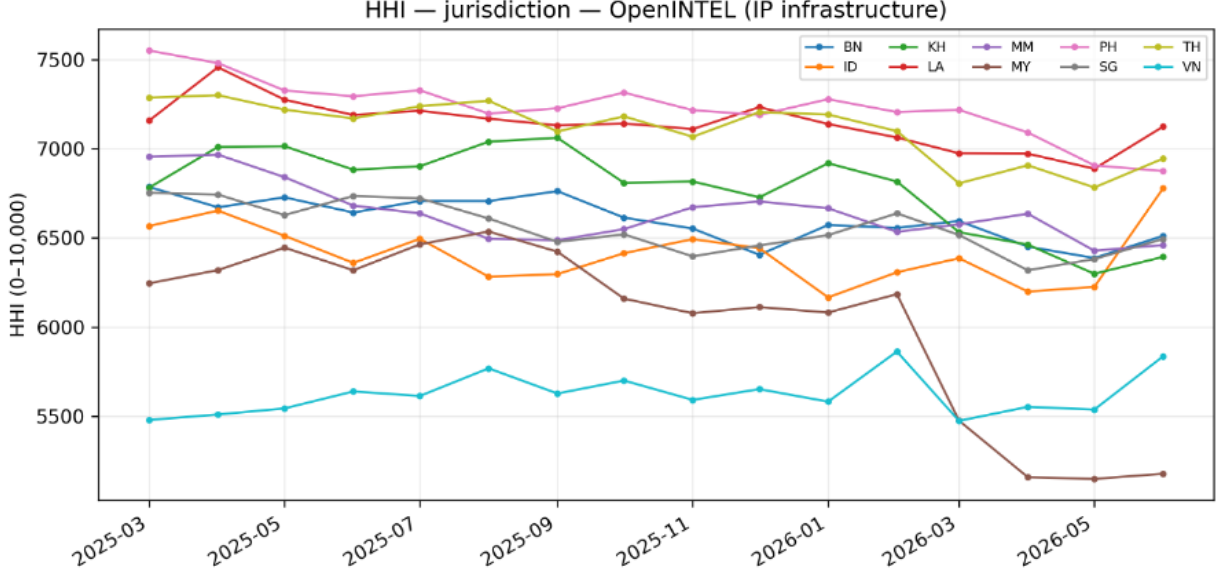


Figure 12 shows the HHI of jurisdictional concentration in the OpenINTEL IP infrastructure data, where values range from 5,150 to 7,550. A total of nine of 10 Southeast Asian countries remain extremely concentrated in almost every month. Malaysia falls from 6,180 in February 2026 to 5,150 in April 2026 and ends below Vietnam in the second quarter of 2026 (Appendix 6).

Figure 12: HHI of jurisdiction concentration, OpenINTEL (IP infrastructure), per country over time.



Chinese technology investment and Chinese market share (RQ3)

Table 1 reports the fixed-effects and random-effects estimates for Model 1a. Here, the Chinese market share of the CDN provision across Southeast Asia measured from Nedko data is regressed on the lagged count of Chinese technology projects, their lagged total value and a lagged construction indicator, over 10 countries and three years. Model 1a itself is not statistically significant (i.e., $F(3,17) = 1.72$ and $p = 0.200$ under fixed-effects). In addition, the Hausman test does not reject random-effects ($\chi^2 = 0.92$, $p = 0.822$). We can see that, from the random-effects model, an additional billion US dollars of Chinese technology investment in the previous year is associated with a drop of around 0.2 percentage points in the Chinese market share in CDN provision ($\beta = -0.0002$, $p < 0.001$).

Table 1: Model 1a — Chinese CDN market concentration (*CDN_China*), Nedko

Variable	FE coefficient	FE S.E.	RE coefficient	RE S.E.
<i>CGIT_projects_{i,t-1}</i>	-0.0979	0.0773	-0.0617	0.0700
<i>CGIT_amount_{i,t-1}</i>	-0.0002***	0.0000	-0.0002***	0.0000
<i>CGIT_type_{i,t-1}</i>	0.1636	0.1160	0.0926	0.1059
constant	(absorbed)		0.7779***	0.0839
N (observations)	30		30	
N (groups)	10		10	
R ² (within)	0.2331		0.2199	
R ² (overall)	-0.2229		-0.0630	
F-statistic / Wald	F(3,17)=1.72,		F(3,26)=1.51,	
χ^2	$p=0.200$		$p=0.235$	

Hausman test (FE vs RE): $\chi^2(3)=0.92$, $p=0.822$.

Panel: 10 countries $\times$ T=3, N=30.

Cluster-robust SEs by country.

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table 2 reports Model 1b, which applies the same specification as Model 1a, with the dependent variable measured from the ISOC data. Again, the Hausman test does not reject random-effects ($\chi^2 = 0.08$, $p=0.994$). No explanatory variable sourced from the CGIT data is associated with the dependent variable measured from the ISOC data at any conventional level of statistical significance.

Table 2: Model 1b — Chinese CDN market concentration (*CDN_China*), ISOC source

Variable	FE coefficient	FE S.E.	RE coefficient	RE S.E.
<i>CGIT_projects_{i,t-1}</i>	-0.2261	0.1798	-0.2036	0.1240
<i>CGIT_amount_{i,t-1}</i>	0.0004	0.0002	0.0004	0.0002
<i>CGIT_type_{i,t-1}</i>	-0.2091	0.3314	-0.2591	0.2971
constant	(absorbed)		1.6777***	0.2846
N (observations)	30		30	
N (Groups)	10		10	
R ² (within)	0.1058		0.1046	
R ² (overall)	-0.1109		0.0446	
F-statistic / Wald	F(3,17)=0.67,		F(3,26)=0.81,	
χ^2	$p=0.582$		$p=0.502$	

Hausman test (FE vs RE): $\chi^2(3)=0.08$, $p=0.994$.

Panel: 10 countries $\times$ T=3, N=30.

Cluster-robust SEs by country.

Significance: *** $p<0.001$, ** $p<0.01$, * $p<0.05$.

Table 3 reports the fixed-effects and random-effects estimates for Model 2, in which the dependent variable is the Chinese market share of IP infrastructure measured from OpenINTEL data over 10 countries and two years. The Hausman statistic is -2.20 which is beyond the supporting range of the χ^2 distribution. Therefore, the Hausman test is uninformative in Table 3 and both fixed-effects and random-effects are reported without statistical preference. We can see that a country-year with at least one Chinese construction transaction in the previous year is associated with a Chinese market share in IP infrastructure provision in the corresponding Southeast Asian market at around 3.7 percentage points lower under fixed-effects and some 3.1 percentage points lower under random-effects ($\beta = -3.6681$ and $\beta = -3.1363$ respectively, both $p < 0.001$).

Table 3: Model 2 — Chinese IP-infrastructure market concentration (*IP_China*), OpenINTEL

Variable	FE coefficient	FE S.E.	RE coefficient	RE S.E.
<i>CGIT_projects_{i,t-1}</i>	-0.0280	0.1266	0.0917	0.1475
<i>CGIT_amount_{i,t-1}</i>	0.0007	0.0007	-0.0009	0.0006
<i>CGIT_type_{i,t-1}</i>	-3.6681***	0.1261	-3.1363***	0.2173
constant	(absorbed)		1.9437**	0.5048
N (observations)	20		20	
N (groups)	10		10	
R ² (within)	0.8329		0.8139	
R ² (overall)	-0.2318		-0.2136	
F-statistic / Wald	F(3,7)=11.63,		F(3,16)=7.09,	
χ^2	$p=0.004$		$p=0.003$	

Hausman test (FE vs RE): $\chi^2(3)=-2.20$, $p=1.000$.

Panel: 10 countries $\times$ T=2, N=20.

Cluster-robust SEs by country.

Significance: *** $p<0.001$, ** $p<0.01$, * $p<0.05$.

Discussion

Existing academic, policy and media debates on the DSR concentrate on Chinese penetration of Global South digital infrastructure (e.g., Heeks et al., 2024; Nouwens, 2021; Thussu, 2025). However, in terms of Chinese penetration of Internet infrastructure provision, we found that Chinese jurisdictional providers, in both CDN and IP infrastructure, hold only marginal market shares. The Chinese CDN market share never exceeds 3.9 per cent in any country-year in either the Nedko or ISOC datasets. The Chinese IP infrastructure market share, additionally, does not exceed 4.8 per cent in any country-year. Meanwhile, the American market share in CDN provision ranges between 95.9 and 99.6 per cent in every country-year we measured. Our findings indicate that the Southeast Asian markets are extremely highly concentrated by American providers.

In CDN provision, we found the Southeast Asian markets to be jurisdictionally saturated. The jurisdictional Gini coefficients nearly reach the statistical ceiling (i.e., at 0.667), while the jurisdictional HHI scores sit between 8,950 and 10,000. In IP infrastructure market share, we found that jurisdictional HHI scores range from around 5,150 to 7,550, where the non-American and non-Chinese market share ranges between 14.2 and 30.8 per cent. We conclude that content delivery is supplied by a small number of global, especially American, providers, whereas IP infrastructure includes national and regional networks, hosting companies and transit providers with a local presence. Therefore, a research study measuring CDN provision alone would report near-total foreign dependency, while a study measuring IP infrastructure provision exclusively would report a more diverse and less concentrated market across Southeast Asia.

Our provider-level empirical findings reinforce the concern about single points of failure raised in our introduction (Mantellassi & Paoli, 2024; Wilt, 2025). We found that Cloudflare leads in every country and in every dataset, holding between 42.9 and 83.0 per cent of the Southeast Asian market share in 2026. We further found that our provider-level HHI scores reach approximately 5,460 in the Nedko dataset and approximately 7,280 in the ISOC dataset, indicating an extreme level of market concentration. The underlying implication is that a disruption at one company, such as Cloudflare, would severely disrupt the provision of Internet infrastructure services across all 10 Southeast Asian countries. Overall, the Gini coefficients and HHI scores presented help answer RQ1 and RQ2.

Compared to the Gini coefficients and HHI scores (in Figures 1 to 12), the panel regression models (in Tables 1 to 3) present limited statistically meaningful information. This is not caused by any flaw in model specification. The limitations exhibited in Tables 1 to 3 are expected, given the thinness of the data extracted from the CGIT database. Ideally, we would rely on the IISS China Connects database to collect extensive global panel data on DSR activities. However, the IISS China Connects database only has available data until 2020. Therefore, it is infeasible to source the explanatory variables of Models 1a, 1b and 2 from the IISS China Connects database. The only publicly available counterpart that we could find as a proxy for the IISS China Connects database is the CGIT database, where we exclusively extracted data within the technology sector. Relative

to the IISS China Connects database, CGIT data are thin. We acknowledge the data limitation when choosing the CGIT database as a proxy, which results in a lack of statistically meaningful outcomes from Tables 1 to 3. It is noteworthy that the CGIT database only provides data on China-Globe bilateral agreements in the technology sector. These agreements alone do not fully represent all DSR activities. Therefore, CGIT data are used only as a proxy. The IISS China Connects database would have offered richer data, as it records DSR-related activities of all kinds.

Among all 10 Southeast Asian markets, we found Malaysia's market share trajectory to be particularly interesting, given its volatility. In IP infrastructure provision in Malaysia, the Chinese market share rose from 1.3 per cent in 2025 to 4.8 per cent in 2026, while the American share fell from 76.1 to 69.5 over the same period. Despite the notable volatility, our findings clearly indicate that American providers remain dominant in the Malaysian market.

To conclude, we reiterate two limitations in this research study. First, to answer RQ1, for both the Nedko and ISOC CDN datasets, we relied on manual Google searches to locate each provider's website and to identify its legal name. We then located the corresponding jurisdiction of each provider by searching the legal name on the OpenCorporates database. However, we cannot fully guarantee that the search results are the actual websites of the CDN providers we intended to identify. Second, for the OpenINTEL IP infrastructure datasets, we programmatically took the country in which an organisation registered its network with a regional Internet registry as the jurisdiction of the provider. The mapping outcomes are proxies for jurisdiction. Technically speaking, an organisation may register its network in one country while the controlling entity belongs to another country. These two limitations should be taken into account by researchers seeking to replicate this study.

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Appendices

Appendix 1: Market share (%) by provider $\times$ year $\times$ country (Nedko (CDN)). Cells = mean monthly share in that year.

Country	Provider	2024	2025	2026
BN	CloudFlare	53.5	55.9	58.6
	Netlify	31.9	31.0	29.2
	Amazon CloudFront	6.5	5.6	5.0
	Fastly	3.4	2.6	2.3
	Imperva	0.6	0.5	0.5
	CDN77	0.7	1.1	1.5
	Akamai	0.6	0.4	0.5
	Alibaba Cloud CDN	0.7	0.6	0.4
	Azure CDN	1.1	1.3	1.0
	BunnyCDN	0.2	0.3	0.3
	Google Cloud CDN	0.3	0.2	0.2
	Vercel	0.2	0.3	0.3
	Tencent Cloud CDN	0.0	0.0	–
	Sucuri	0.3	0.2	0.2
	SiteGround CDN	–	–	–
ID	CloudFlare	53.9	59.2	66.5
	Netlify	27.4	24.0	20.1
	Amazon CloudFront	14.0	11.2	8.1
	Fastly	1.5	1.7	1.4
	Imperva	0.3	0.2	0.1
	CDN77	0.6	0.8	1.0
	Akamai	0.6	0.6	0.6
	Alibaba Cloud CDN	0.6	0.8	0.8
	Azure CDN	0.1	0.1	0.1
	BunnyCDN	0.2	0.4	0.4
	Google Cloud CDN	0.3	0.6	0.5
	Vercel	0.2	0.1	0.1
	Tencent Cloud CDN	–	0.0	0.1
	Sucuri	0.1	0.2	0.1
	SiteGround CDN	–	–	–
KH	CloudFlare	55.4	56.4	59.9
	Netlify	33.7	32.4	29.8

	Amazon CloudFront	5.0	4.6	3.3
	Fastly	1.9	2.4	2.6
	Imperva	0.4	0.2	0.0
	CDN77	0.7	1.2	1.5
	Akamai	0.9	0.7	0.6
	Alibaba Cloud CDN	0.6	0.5	0.5
	Azure CDN	0.4	0.3	0.4
	BunnyCDN	0.3	0.3	0.6
	Google Cloud CDN	0.2	0.1	0.1
	Vercel	0.1	0.3	0.4
	Tencent Cloud CDN	0.2	0.2	0.2
	Sucuri	0.1	0.1	0.1
	SiteGround CDN	0.2	0.1	0.1
LA	CloudFlare	62.1	65.7	67.4
	Netlify	28.1	25.0	22.6
	Amazon CloudFront	5.0	4.4	4.0
	Fastly	1.3	0.9	1.1
	Imperva	0.5	0.2	0.1
	CDN77	0.3	1.0	2.1
	Akamai	0.8	0.7	0.8
	Alibaba Cloud CDN	0.3	0.5	0.3
	Azure CDN	0.5	0.2	0.2
	BunnyCDN	0.4	0.3	0.4
	Google Cloud CDN	0.4	0.3	0.1
	Vercel	0.1	0.3	0.5
	Tencent Cloud CDN	0.2	0.3	0.2
	Sucuri	0.0	0.1	0.0
	SiteGround CDN	–	–	–
MM	CloudFlare	61.1	62.8	62.7
	Netlify	26.3	25.7	26.6
	Amazon CloudFront	6.1	5.0	3.5
	Fastly	2.3	2.0	2.5
	Imperva	0.9	1.1	0.6
	CDN77	0.4	1.1	1.4
	Akamai	0.9	0.8	0.6
	Alibaba Cloud CDN	0.6	0.5	0.7
	Azure CDN	0.3	0.3	0.5
	BunnyCDN	0.4	0.4	0.4
	Google Cloud CDN	0.5	0.2	0.2
	Vercel	0.1	0.2	0.2
	Tencent Cloud CDN	0.0	0.0	–
	Sucuri	–	–	–
	SiteGround CDN	–	–	–
MY	CloudFlare	52.7	54.0	53.5
	Netlify	31.7	30.1	31.0
	Amazon CloudFront	6.2	6.4	6.1
	Fastly	2.9	2.7	2.8

	Imperva	2.8	2.6	2.5
	CDN77	1.1	1.2	1.3
	Akamai	0.7	0.4	0.4
	Alibaba Cloud CDN	0.3	0.4	0.5
	Azure CDN	0.7	0.4	0.3
	BunnyCDN	0.2	0.7	0.8
	Google Cloud CDN	0.5	0.6	0.4
	Vercel	0.1	0.2	0.3
	Tencent Cloud CDN	0.1	0.1	0.1
	Sucuri	0.0	0.0	0.1
	SiteGround CDN	–	–	–
PH	CloudFlare	50.0	55.8	62.7
	Netlify	31.6	28.4	22.8
	Amazon CloudFront	10.7	9.1	8.5
	Fastly	2.9	2.4	2.2
	Imperva	0.7	0.2	0.4
	CDN77	0.8	1.2	1.4
	Akamai	1.1	0.7	0.5
	Alibaba Cloud CDN	0.7	0.9	0.8
	Azure CDN	0.7	0.4	0.2
	BunnyCDN	0.1	0.1	0.2
	Google Cloud CDN	0.3	0.3	0.1
	Vercel	0.1	0.3	0.2
	Tencent Cloud CDN	–	–	–
	Sucuri	0.3	0.1	0.0
	SiteGround CDN	0.1	–	–
SG	CloudFlare	44.3	46.5	48.9
	Netlify	31.3	30.3	28.5
	Amazon CloudFront	12.8	11.9	10.9
	Fastly	4.0	3.7	3.6
	Imperva	2.8	2.7	3.4
	CDN77	1.0	1.2	1.3
	Akamai	1.0	0.8	0.7
	Alibaba Cloud CDN	1.2	1.2	1.1
	Azure CDN	0.7	0.6	0.6
	BunnyCDN	0.2	0.2	0.3
	Google Cloud CDN	0.1	0.1	–
	Vercel	0.2	0.3	0.4
	Tencent Cloud CDN	0.3	0.3	0.3
	Sucuri	0.0	0.0	0.0
	SiteGround CDN	–	–	–
TH	CloudFlare	57.3	56.1	61.9
	Netlify	28.2	29.2	26.8
	Amazon CloudFront	6.2	6.7	5.5
	Fastly	1.3	1.4	0.9
	Imperva	2.8	2.3	1.2
	CDN77	1.0	1.3	1.3

	Akamai	0.7	0.6	0.5
	Alibaba Cloud CDN	0.6	0.5	0.4
	Azure CDN	0.8	0.7	0.3
	BunnyCDN	0.4	0.6	0.7
	Google Cloud CDN	0.2	0.1	–
	Vercel	0.2	0.3	0.4
	Tencent Cloud CDN	0.2	0.2	0.1
	Sucuri	–	0.0	0.1
	SiteGround CDN	–	–	–
VN	CloudFlare	45.5	52.4	57.6
	Netlify	43.6	36.5	29.4
	Amazon CloudFront	7.5	7.9	9.4
	Fastly	1.1	0.9	0.8
	Imperva	0.2	0.1	0.3
	CDN77	0.7	0.8	0.9
	Akamai	0.5	0.4	0.3
	Alibaba Cloud CDN	0.2	0.3	0.6
	Azure CDN	0.5	0.4	0.2
	BunnyCDN	0.2	0.1	0.1
	Google Cloud CDN	0.1	0.1	0.2
	Vercel	0.1	0.1	0.1
	Tencent Cloud CDN	–	–	–
	Sucuri	–	–	–
	SiteGround CDN	–	–	–

Appendix 2: Market share (%) by provider $\times$ year $\times$ country (ISOC (CDN)). Cells = mean monthly share in that year.

Country	Provider	2024	2025	2026
BN	cloudflare	65.2	74.4	75.0
	amazon	7.7	9.6	8.5
	akamai	16.7	7.4	8.0
	fastly	4.0	4.0	4.0
	alibaba	0.4	0.9	1.0
	microsoft	4.5	2.5	2.1
	facebook	0.7	1.0	1.0
	cloudfront	0.7	–	–
	cdn77	0.0	0.1	0.4
	gcore	0.0	0.1	–
	bunnycdn	0.0	–	–
	netflix	–	0.0	–
	wordpress	–	0.0	–
	edgio	–	–	–
ID	cloudflare	73.1	76.4	77.9
	amazon	17.2	15.0	13.2
	akamai	3.6	3.6	3.7
	fastly	1.4	1.6	1.2

	alibaba	1.1	1.4	1.5
	microsoft	0.7	0.5	0.4
	facebook	1.2	1.1	1.0
	cloudfront	1.8	–	–
	cdn77	0.1	0.3	0.8
	gcore	–	–	–
	bunnycdn	0.0	0.0	0.3
	netflix	–	–	–
	wordpress	–	–	–
	edgio	–	–	–
KH	cloudflare	73.7	72.6	70.6
	amazon	11.0	12.4	14.2
	akamai	6.8	7.1	6.6
	fastly	2.4	3.0	3.2
	alibaba	2.2	2.4	2.4
	microsoft	1.0	0.7	0.9
	facebook	1.7	1.7	1.6
	cloudfront	1.2	–	–
	cdn77	0.1	0.1	0.5
	gcore	–	–	–
	bunnycdn	–	–	–
	netflix	–	–	–
	wordpress	0.0	–	–
	edgio	–	–	–
LA	cloudflare	83.3	83.6	83.0
	amazon	6.4	7.1	7.3
	akamai	5.2	5.0	5.5
	fastly	1.2	1.0	1.0
	alibaba	0.6	0.9	0.7
	microsoft	1.0	0.7	0.7
	facebook	1.4	1.5	1.4
	cloudfront	0.8	–	–
	cdn77	–	0.1	0.4
	gcore	–	0.1	–
	bunnycdn	–	0.0	–
	netflix	–	0.0	–
	wordpress	–	–	–
	edgio	–	–	–
MM	cloudflare	73.0	75.5	78.8
	amazon	12.2	10.8	8.4
	akamai	5.3	5.9	4.7
	fastly	2.4	2.3	3.6
	alibaba	3.9	3.4	2.1
	microsoft	1.3	1.5	1.7
	facebook	0.3	0.5	0.5
	cloudfront	1.4	–	–
	cdn77	0.1	0.1	0.1

	gcore	—	0.0	—
	bunnycdn	0.0	—	—
	netflix	—	—	—
	wordpress	—	—	—
	edgio	—	—	—
MY	cloudflare	76.0	74.5	73.2
	amazon	9.1	10.7	12.6
	akamai	7.3	6.7	6.2
	fastly	3.3	3.1	2.3
	alibaba	0.4	1.0	1.4
	microsoft	2.2	2.6	2.4
	facebook	1.0	1.3	1.4
	cloudfront	0.7	—	—
	cdn77	0.0	0.1	0.3
	gcore	—	0.1	—
	bunnycdn	—	—	—
	netflix	—	0.0	0.1
	wordpress	—	—	—
	edgio	—	—	—
PH	cloudflare	62.2	66.5	72.3
	amazon	21.5	18.7	14.7
	akamai	7.3	6.8	5.5
	fastly	3.3	3.0	2.8
	alibaba	1.1	1.7	2.1
	microsoft	1.8	2.0	1.6
	facebook	1.1	1.1	0.9
	cloudfront	1.7	—	—
	cdn77	0.0	0.1	0.1
	gcore	—	—	—
	bunnycdn	—	—	—
	netflix	—	0.0	—
	wordpress	—	—	—
	edgio	—	—	—
SG	cloudflare	57.8	58.8	58.6
	amazon	16.7	17.0	16.6
	akamai	14.1	14.0	13.9
	fastly	5.2	4.6	4.5
	alibaba	1.6	2.2	2.1
	microsoft	2.1	2.2	2.7
	facebook	1.0	1.1	1.3
	cloudfront	1.5	—	—
	cdn77	—	0.1	0.3
	gcore	—	—	—
	bunnycdn	—	—	—
	netflix	—	0.0	—
	wordpress	—	—	—

	edgio	0.0	–	–
TH	cloudflare	79.5	77.9	78.0
	amazon	8.7	8.7	7.3
	akamai	6.2	8.1	8.8
	fastly	1.3	1.0	1.3
	alibaba	1.4	2.0	1.6
	microsoft	1.3	1.2	1.3
	facebook	0.6	0.8	0.8
	cloudfront	0.7	–	–
	cdn77	0.1	0.2	0.6
	gcore	–	0.1	–
	bunnycdn	0.1	0.1	0.1
	netflix	–	0.0	0.1
	wordpress	–	–	–
	edgio	–	–	–
VN	cloudflare	68.5	80.2	81.4
	amazon	9.4	11.8	11.5
	akamai	14.2	3.3	3.1
	fastly	0.6	1.1	1.1
	alibaba	1.1	1.9	1.1
	microsoft	3.7	0.6	0.4
	facebook	1.0	1.1	1.1
	cloudfront	1.2	–	–
	cdn77	0.1	0.0	–
	gcore	0.1	–	–
	bunnycdn	0.0	–	–
	netflix	–	0.0	0.1
	wordpress	0.0	–	–
	edgio	–	–	–

Appendix 3: Market share (%) by provider $\times$ year $\times$ country (OpenINTEL (IP infrastructure)).
Cells = mean monthly share in that year.

Country	Provider	2025	2026
BN	Cloudflare, Inc. (CLOUDFLARENET)	52.8	53.3
	Amazon.com, Inc. (AMAZON-02)	6.5	6.7
	Google LLC (GOOGLE)	3.1	2.8
	Akamai International B.V. (AKAMAI-ASN1)	2.4	2.8
	Akamai Technologies, Inc. (AKAMAI-AS)	3.1	2.7
	Fastly, Inc. (FASTLY)	2.9	2.3
	DataWeb Global Group B.V. (ADVANCEDHOSTERS-AS)	2.5	3.0
	Google LLC (GOOGLE-CLOUD-PLATFORM)	1.3	1.0
	OVH SAS (OVH)	1.9	1.5
	Microsoft Corporation (MICROSOFT-CORP-MSN- AS-BLOCK)	1.6	2.0
	Facebook, Inc. (FACEBOOK)	0.7	0.7

	Incapsula Inc (INCAPSULA)	0.3	0.4
	Other	20.9	20.8
ID	Cloudflare, Inc. (CLOUDFLARENET)	46.1	46.7
	Amazon.com, Inc. (AMAZON-02)	11.5	9.4
	Google LLC (GOOGLE)	4.0	3.5
	Akamai International B.V. (AKAMAI-ASN1)	2.3	2.6
	Akamai Technologies, Inc. (AKAMAI-AS)	0.9	1.0
	Fastly, Inc. (FASTLY)	1.4	1.5
	DataWeb Global Group B.V. (ADVANCEDHOSTERS-AS)	1.4	1.3
	Google LLC (GOOGLE-CLOUD-PLATFORM)	2.7	2.7
	OVH SAS (OVH)	4.6	4.3
	Microsoft Corporation (MICROSOFT-CORP-MSN- AS-BLOCK)	0.5	0.5
	Facebook, Inc. (FACEBOOK)	1.0	1.0
	Incapsula Inc (INCAPSULA)	0.3	0.2
	Other	23.4	25.3
KH	Cloudflare, Inc. (CLOUDFLARENET)	52.4	53.7
	Amazon.com, Inc. (AMAZON-02)	8.8	7.9
	Google LLC (GOOGLE)	4.1	3.7
	Akamai International B.V. (AKAMAI-ASN1)	2.4	2.8
	Akamai Technologies, Inc. (AKAMAI-AS)	2.6	2.3
	Fastly, Inc. (FASTLY)	2.3	2.1
	DataWeb Global Group B.V. (ADVANCEDHOSTERS-AS)	1.8	1.7
	Google LLC (GOOGLE-CLOUD-PLATFORM)	1.2	1.6
	OVH SAS (OVH)	1.0	1.0
	Microsoft Corporation (MICROSOFT-CORP-MSN- AS-BLOCK)	0.5	0.6
	Facebook, Inc. (FACEBOOK)	1.3	1.0
	Incapsula Inc (INCAPSULA)	0.1	–
	Other	21.4	21.5
LA	Cloudflare, Inc. (CLOUDFLARENET)	60.8	60.7
	Amazon.com, Inc. (AMAZON-02)	5.3	5.5
	Google LLC (GOOGLE)	3.3	3.1
	Akamai International B.V. (AKAMAI-ASN1)	1.6	2.1
	Akamai Technologies, Inc. (AKAMAI-AS)	2.0	1.7
	Fastly, Inc. (FASTLY)	0.9	1.0
	DataWeb Global Group B.V. (ADVANCEDHOSTERS-AS)	2.4	2.6
	Google LLC (GOOGLE-CLOUD-PLATFORM)	2.0	2.1
	OVH SAS (OVH)	0.6	0.8
	Microsoft Corporation (MICROSOFT-CORP-MSN- AS-BLOCK)	0.5	0.4
	Facebook, Inc. (FACEBOOK)	1.1	1.1
	Incapsula Inc (INCAPSULA)	0.1	–

	Other	19.4	19.0
MM	Cloudflare, Inc. (CLOUDFLARENET)	54.5	56.9
	Amazon.com, Inc. (AMAZON-02)	7.5	5.8
	Google LLC (GOOGLE)	3.0	3.3
	Akamai International B.V. (AKAMAI-ASN1)	2.2	2.0
	Akamai Technologies, Inc. (AKAMAI-AS)	1.7	1.3
	Fastly, Inc. (FASTLY)	2.0	2.4
	DataWeb Global Group B.V. (ADVANCEDHOSTERS-AS)	2.2	2.2
	Google LLC (GOOGLE-CLOUD-PLATFORM)	1.4	1.2
	OVH SAS (OVH)	0.9	0.7
	Microsoft Corporation (MICROSOFT-CORP-MSN- AS-BLOCK)	1.1	1.2
	Facebook, Inc. (FACEBOOK)	0.4	0.4
	Incapsula Inc (INCAPSULA)	1.2	0.6
	Other	21.8	22.0
MY	Cloudflare, Inc. (CLOUDFLARENET)	49.8	43.9
	Amazon.com, Inc. (AMAZON-02)	7.6	7.6
	Google LLC (GOOGLE)	3.0	3.0
	Akamai International B.V. (AKAMAI-ASN1)	2.3	2.6
	Akamai Technologies, Inc. (AKAMAI-AS)	2.0	1.9
	Fastly, Inc. (FASTLY)	2.1	2.0
	DataWeb Global Group B.V. (ADVANCEDHOSTERS-AS)	2.8	2.4
	Google LLC (GOOGLE-CLOUD-PLATFORM)	1.1	1.2
	OVH SAS (OVH)	1.3	0.8
	Microsoft Corporation (MICROSOFT-CORP-MSN- AS-BLOCK)	1.3	1.3
	Facebook, Inc. (FACEBOOK)	0.9	0.9
	Incapsula Inc (INCAPSULA)	2.1	1.9
	Other	23.5	30.4
PH	Cloudflare, Inc. (CLOUDFLARENET)	50.8	52.4
	Amazon.com, Inc. (AMAZON-02)	14.1	12.5
	Google LLC (GOOGLE)	3.1	3.1
	Akamai International B.V. (AKAMAI-ASN1)	2.5	2.5
	Akamai Technologies, Inc. (AKAMAI-AS)	2.6	2.7
	Fastly, Inc. (FASTLY)	2.4	2.3
	DataWeb Global Group B.V. (ADVANCEDHOSTERS-AS)	1.0	1.3
	Google LLC (GOOGLE-CLOUD-PLATFORM)	1.8	1.5
	OVH SAS (OVH)	0.9	0.9
	Microsoft Corporation (MICROSOFT-CORP-MSN- AS-BLOCK)	1.6	1.2
	Facebook, Inc. (FACEBOOK)	0.9	0.9
	Incapsula Inc (INCAPSULA)	0.1	0.2
	Other	18.1	18.5

SG	Cloudflare, Inc. (CLOUDFLARENET)	42.4	42.9
	Amazon.com, Inc. (AMAZON-02)	12.4	12.2
	Google LLC (GOOGLE)	3.2	3.3
	Akamai International B.V. (AKAMAI-ASN1)	5.3	5.2
	Akamai Technologies, Inc. (AKAMAI-AS)	5.2	4.2
	Fastly, Inc. (FASTLY)	3.6	3.2
	DataWeb Global Group B.V. (ADVANCEDHOSTERS-AS)	2.1	1.9
	Google LLC (GOOGLE-CLOUD-PLATFORM)	0.9	0.8
	OVH SAS (OVH)	0.9	0.8
	Microsoft Corporation (MICROSOFT-CORP-MSN- AS-BLOCK)	1.7	2.0
	Facebook, Inc. (FACEBOOK)	0.9	0.8
	Incapsula Inc (INCAPSULA)	2.3	2.4
	Other	19.2	20.4
TH	Cloudflare, Inc. (CLOUDFLARENET)	57.6	56.6
	Amazon.com, Inc. (AMAZON-02)	6.7	6.2
	Google LLC (GOOGLE)	2.7	3.0
	Akamai International B.V. (AKAMAI-ASN1)	1.1	1.7
	Akamai Technologies, Inc. (AKAMAI-AS)	1.6	1.3
	Fastly, Inc. (FASTLY)	1.0	0.8
	DataWeb Global Group B.V. (ADVANCEDHOSTERS-AS)	1.1	1.3
	Google LLC (GOOGLE-CLOUD-PLATFORM)	0.8	0.8
	OVH SAS (OVH)	0.9	1.0
	Microsoft Corporation (MICROSOFT-CORP-MSN- AS-BLOCK)	0.9	1.1
	Facebook, Inc. (FACEBOOK)	0.6	0.6
	Incapsula Inc (INCAPSULA)	1.7	1.1
	Other	23.3	24.4
VN	Cloudflare, Inc. (CLOUDFLARENET)	46.4	45.2
	Amazon.com, Inc. (AMAZON-02)	8.3	9.8
	Google LLC (GOOGLE)	2.9	3.3
	Akamai International B.V. (AKAMAI-ASN1)	1.2	1.5
	Akamai Technologies, Inc. (AKAMAI-AS)	1.2	1.0
	Fastly, Inc. (FASTLY)	1.0	1.1
	DataWeb Global Group B.V. (ADVANCEDHOSTERS-AS)	0.9	0.9
	Google LLC (GOOGLE-CLOUD-PLATFORM)	1.0	0.8
	OVH SAS (OVH)	0.6	0.6
	Microsoft Corporation (MICROSOFT-CORP-MSN- AS-BLOCK)	0.4	0.4
	Facebook, Inc. (FACEBOOK)	0.8	0.9
	Incapsula Inc (INCAPSULA)	0.1	0.1
	Other	35.0	34.5

Appendix 4: Market share (%) by jurisdiction $\times$ year $\times$ country (Nedko (CDN)). Cells = mean monthly share in that year.

Country	Jurisdiction	2024	2025	2026
BN	China	0.7	0.6	0.4
	USA	98.3	98.1	97.7
	Other	1.0	1.4	1.9
ID	China	0.6	0.9	0.8
	USA	98.5	98.0	97.8
	Other	0.8	1.2	1.4
KH	China	0.8	0.7	0.7
	USA	98.1	97.6	97.1
	Other	1.2	1.7	2.1
LA	China	0.6	0.7	0.6
	USA	98.7	97.9	96.9
	Other	0.7	1.4	2.5
MM	China	0.6	0.5	0.7
	USA	98.5	98.0	97.5
	Other	0.9	1.5	1.8
MY	China	0.4	0.6	0.6
	USA	98.3	97.6	97.3
	Other	1.3	1.9	2.1
PH	China	0.7	0.9	0.8
	USA	98.3	97.8	97.7
	Other	1.1	1.3	1.6
SG	China	1.5	1.6	1.4
	USA	97.3	97.0	97.1
	Other	1.2	1.4	1.5
TH	China	0.8	0.7	0.5
	USA	97.8	97.4	97.5
	Other	1.4	1.9	2.0
VN	China	0.2	0.3	0.6
	USA	99.0	98.7	98.4
	Other	0.8	0.9	1.0

Appendix 5: Market share (%) by jurisdiction $\times$ year $\times$ country (ISOC (CDN)). Cells = mean monthly share in that year.

Country	Jurisdiction	2024	2025	2026
BN	China	0.4	0.9	1.0
	USA	99.5	98.9	98.6
	Other	0.1	0.2	0.4
ID	China	1.1	1.4	1.5
	USA	98.9	98.3	97.4
	Other	0.1	0.3	1.1
KH	China	2.2	2.4	2.4

	USA	97.7	97.5	97.1
	Other	0.1	0.1	0.5
LA	China	0.6	0.9	0.7
	USA	99.4	99.0	99.0
	Other	–	0.2	0.4
MM	China	3.9	3.4	2.1
	USA	95.9	96.5	97.7
	Other	0.1	0.1	0.1
MY	China	0.4	1.0	1.4
	USA	99.6	98.8	98.3
	Other	0.0	0.2	0.3
PH	China	1.1	1.7	2.1
	USA	98.9	98.2	97.8
	Other	0.0	0.1	0.1
SG	China	1.6	2.2	2.1
	USA	98.4	97.8	97.6
	Other	–	0.1	0.3
TH	China	1.4	2.0	1.6
	USA	98.3	97.7	97.6
	Other	0.3	0.4	0.8
VN	China	1.1	1.9	1.1
	USA	98.7	98.1	98.9
	Other	0.2	0.0	–

Appendix 6: Market share (%) by jurisdiction $\times$ year $\times$ country (OpenINTEL (IP infrastructure)). Cells = mean monthly share in that year.

Country	Jurisdiction	2025	2026
BN	China	0.8	0.6
	USA	79.0	77.7
	Other	20.2	21.7
ID	China	1.4	1.4
	USA	77.5	76.4
	Other	21.2	22.2
KH	China	1.4	2.7
	USA	81.3	78.9
	Other	17.3	18.4
LA	China	0.6	0.4
	USA	83.4	81.9
	Other	16.1	17.7
MM	China	2.0	2.4
	USA	79.7	78.6
	Other	18.2	19.0
MY	China	1.3	4.8
	USA	76.1	69.5

	Other	22.5	25.8
PH	China	1.5	1.4
	USA	84.3	82.7
	Other	14.2	15.9
SG	China	3.7	3.2
	USA	79.4	78.2
	Other	16.9	18.5
TH	China	1.5	0.8
	USA	83.5	81.5
	Other	15.0	17.7
VN	China	0.9	0.7
	USA	68.3	68.4
	Other	30.8	30.8

Supplementary Material(s)

The full Python-based pipeline, along with the datasets built and used and the empirical findings visualised, is publicly deposited on a public GitHub repository at <https://github.com/newlivehung123123/pulse>. We made the repository public to optimise research transparency, replicability and reproducibility.